

MATH1560J Assignment 7 - Sample Answer

Exercise 7.1

(a)

$$\cos x = 2 - \cos x \implies 2 \cos x = 2 \implies \cos x = 1 \implies x = 0, 2\pi$$

On the interval $[0, 2\pi]$, $2 - \cos x \geq \cos x$. The area A is:

$$\begin{aligned} A &= \int_0^{2\pi} [(2 - \cos x) - \cos x] dx = \int_0^{2\pi} (2 - 2 \cos x) dx \\ &= [2x - 2 \sin x]_0^{2\pi} = (4\pi - 0) - (0 - 0) = 4\pi \end{aligned}$$

(b)

$$2y^2 = 4 + y^2 \implies y^2 = 4 \implies y = \pm 2$$

The area A is:

$$\begin{aligned} A &= \int_{-2}^2 [(4 + y^2) - 2y^2] dy = \int_{-2}^2 (4 - y^2) dy \\ &= \left[4y - \frac{1}{3}y^3 \right]_{-2}^2 = \left(8 - \frac{8}{3} \right) - \left(-8 + \frac{8}{3} \right) = \frac{16}{3} + \frac{16}{3} = \frac{32}{3} \end{aligned}$$

Exercise 7.2

(a)

Intersection: $\sin x = \cos 2x \implies \sin x = 1 - 2 \sin^2 x \implies (2 \sin x - 1)(\sin x + 1) = 0$.

In $[0, \pi/2]$, $\sin x = 1/2 \implies x = \pi/6$.

$$\begin{aligned} I &= \int_0^{\pi/6} (\cos 2x - \sin x) dx + \int_{\pi/6}^{\pi/2} (\sin x - \cos 2x) dx \\ &= \left[\frac{1}{2} \sin 2x + \cos x \right]_0^{\pi/6} + \left[-\cos x - \frac{1}{2} \sin 2x \right]_{\pi/6}^{\pi/2} \\ &= \left(\frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{2} - 1 \right) + \left(0 - 0 - \left(-\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{4} \right) \right) = \frac{3\sqrt{3}}{2} - 1 \end{aligned}$$

(b)

$$\begin{aligned} I &= \int_{-1}^0 (2^x - 3^x) dx + \int_0^1 (3^x - 2^x) dx \\ &= \left[\frac{2^x}{\ln 2} - \frac{3^x}{\ln 3} \right]_{-1}^0 + \left[\frac{3^x}{\ln 3} - \frac{2^x}{\ln 2} \right]_0^1 \\ &= \left(\frac{1}{\ln 2} - \frac{1}{\ln 3} - \frac{1}{2 \ln 2} + \frac{1}{3 \ln 3} \right) + \left(\frac{3}{\ln 3} - \frac{2}{\ln 2} - \frac{1}{\ln 3} + \frac{1}{\ln 2} \right) \\ &= \frac{4}{3 \ln 3} - \frac{1}{2 \ln 2} \end{aligned}$$

Exercise 7.3

Consider cylinders $x^2 + z^2 = r^2$ and $y^2 + z^2 = r^2$. A horizontal cross-section at height z is a square with side length $s = 2\sqrt{r^2 - z^2}$.

$$V = \int_{-r}^r A(z) dz = \int_{-r}^r [2\sqrt{r^2 - z^2}]^2 dz = 8 \int_0^r (r^2 - z^2) dz = 8 \left[r^2 z - \frac{1}{3} z^3 \right]_0^r = 8 \left(r^3 - \frac{1}{3} r^3 \right) = \frac{16}{3} r^3$$

Exercise 7.4

(a)

Using cylindrical shells:

$$V = \int_0^4 2\pi y(4y^2 - y^3) dy = 2\pi \int_0^4 (4y^3 - y^4) dy = 2\pi \left[y^4 - \frac{1}{5} y^5 \right]_0^4 = 2\pi \left(256 - \frac{1024}{5} \right) = \frac{512\pi}{5}$$

(b)

Radius $r = (2 - x)$, height $h = x^4$:

$$V = \int_0^1 2\pi(2 - x)x^4 dx = 2\pi \int_0^1 (2x^4 - x^5) dx = 2\pi \left[\frac{2}{5} x^5 - \frac{1}{6} x^6 \right]_0^1 = 2\pi \left(\frac{2}{5} - \frac{1}{6} \right) = \frac{7\pi}{15}$$

Exercise 7.5

(a)

Since f is continuous on $[1, 3]$, the Mean Value Theorem for Integrals states:

$$\exists c \in [1, 3] \text{ such that } \int_1^3 f(x) dx = f(c)(3 - 1) \Rightarrow f(c) = 4$$

Thus, there exists $c \in [1, 3]$ such that $f(c) = 4$.

(b)

$$3 = \frac{1}{b} \int_0^b (2 + 6x - 3x^2) dx = \frac{1}{b} [2x + 3x^2 - x^3]_0^b = 2 + 3b - b^2$$

$$b^2 - 3b + 1 = 0 \implies b = \frac{3 \pm \sqrt{5}}{2}$$

Exercise 7.6

(a)

$$y' = \tan x$$

$$L = \int_0^{\pi/4} \sqrt{1 + \tan^2 x} dx = \int_0^{\pi/4} \sec x dx = [\ln |\sec x + \tan x|]_0^{\pi/4} = \ln(\sqrt{2} + 1)$$

(b)

$$y' = \sinh 2x$$

$$L = \int_0^1 \sqrt{1 + \sinh^2 2x} dx = \int_0^1 \cosh 2x dx = \left[\frac{1}{2} \sinh 2x \right]_0^1 = \frac{1}{2} \sinh 2$$

Exercise 7.7

$$y' = \frac{1}{\sqrt{1-x^2}} - \frac{x}{\sqrt{1-x^2}} = \frac{1-x}{\sqrt{1-x^2}} = \sqrt{\frac{1-x}{1+x}}$$

$$s(x) = \int_0^x \sqrt{1 + \frac{1-t}{1+t}} dt = \int_0^x \sqrt{\frac{2}{1+t}} dt = \sqrt{2} [2\sqrt{1+t}]_0^x = 2\sqrt{2}(\sqrt{1+x} - 1) \quad (x \in [-1, 1])$$

Exercise 7.8

(a)

$$S = \int_0^2 2\pi x^3 \sqrt{1+9x^4} dx = \frac{2\pi}{36} \int_1^{145} \sqrt{u} du = \frac{\pi}{27} (145\sqrt{145} - 1)$$

(b)

$$x = \frac{1}{9}y^2 + 2 \implies dx/dy = \frac{2}{9}y. \text{ Bounds for } y \text{ are } [0, 6].$$

$$S = \int_0^6 2\pi y \sqrt{1 + \frac{4}{81}y^2} dy = 2\pi \frac{81}{8} \int_1^{25/9} \sqrt{u} du = \frac{81\pi}{4} \left[\frac{2}{3} u^{3/2} \right]_1^{25/9} = 49\pi$$